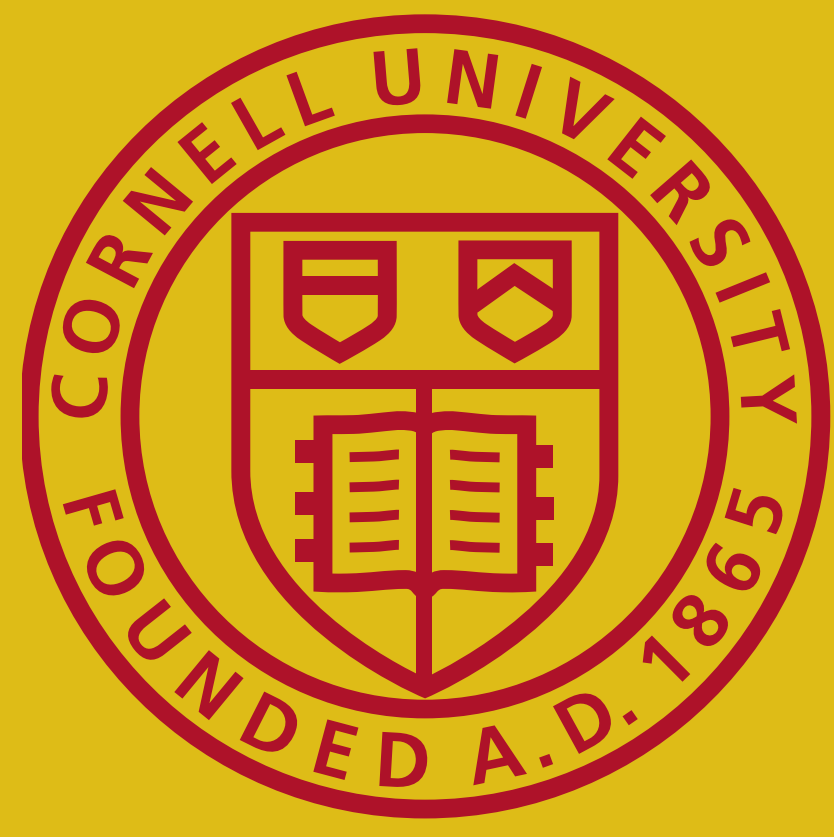




Does overconfidence undermine misinformation interventions?

Jabin Binnendyk¹, Sze Yuh Nina Wang¹, Samantha Phillips², & Gordon Pennycook¹

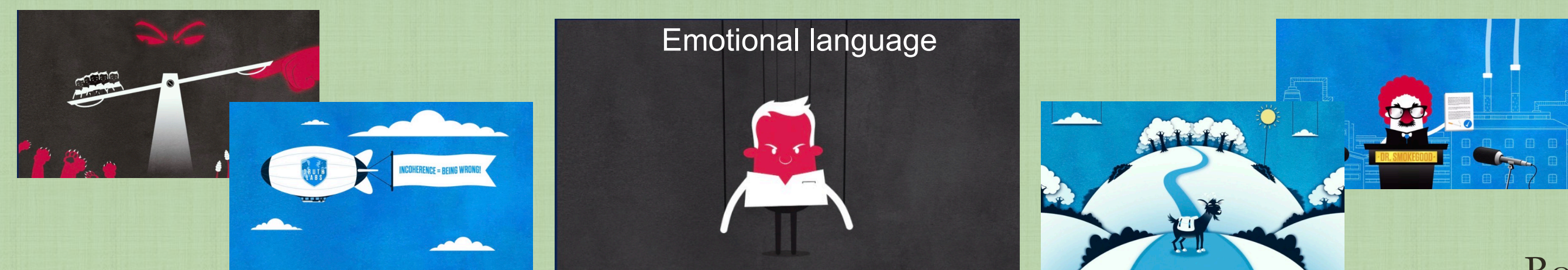
¹ Cornell University, ² Carnegie Mellon University



Commonly used misinformation interventions

1) Psychological inoculation: prebunking approach

Provides tips on spotting falsehoods e.g., emotional language

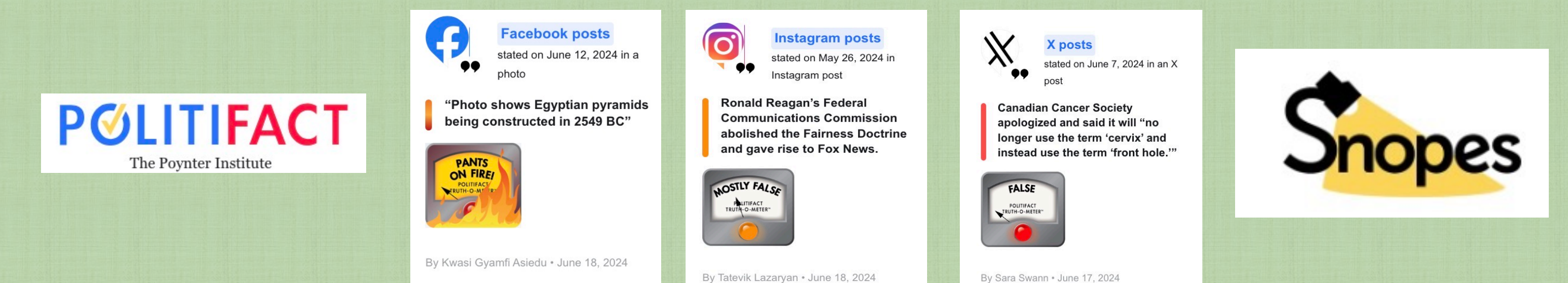


Rozenbeek et al., 2022

Does
overconfidence
moderate their
effectiveness?

2) Fact-checking: debunking approach

participants provided fact-check after misinformation



Psychological Inoculation Study

Intervention (n = 492) / Control (n = 508)

DV = Stimuli ratings (10 items)

Manipulative or neutral – randomized pairs

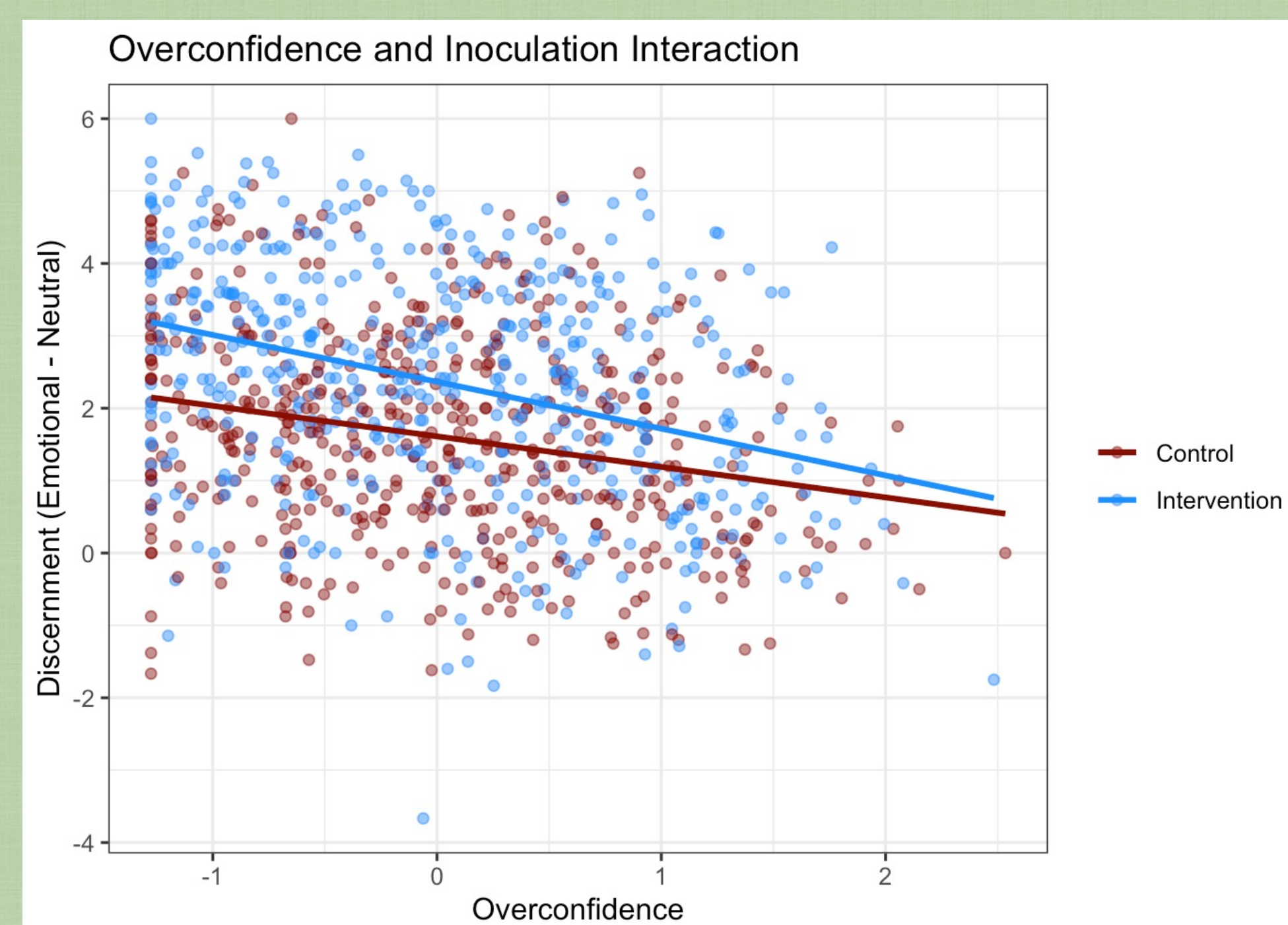
IV = GOT

Discernment ~ GOT * Intervention

GOT [b = -.53, p < .001]

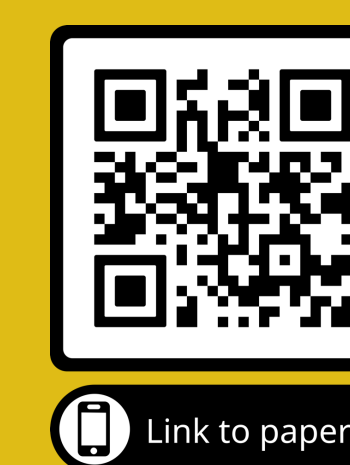
Inoculation [b = .75, p < .001]

GOT : Inoculation [b = -.23, p = .042]



Generalized Overconfidence Task (GOT)

- General tendency to be overconfident



Binnendyk & Pennycook, 2024

Fact-checking Study

Intervention (n = 500) / Control (n = 505)

DV = Average belief (4 posts ranging on topics)

High or low emotionality posts

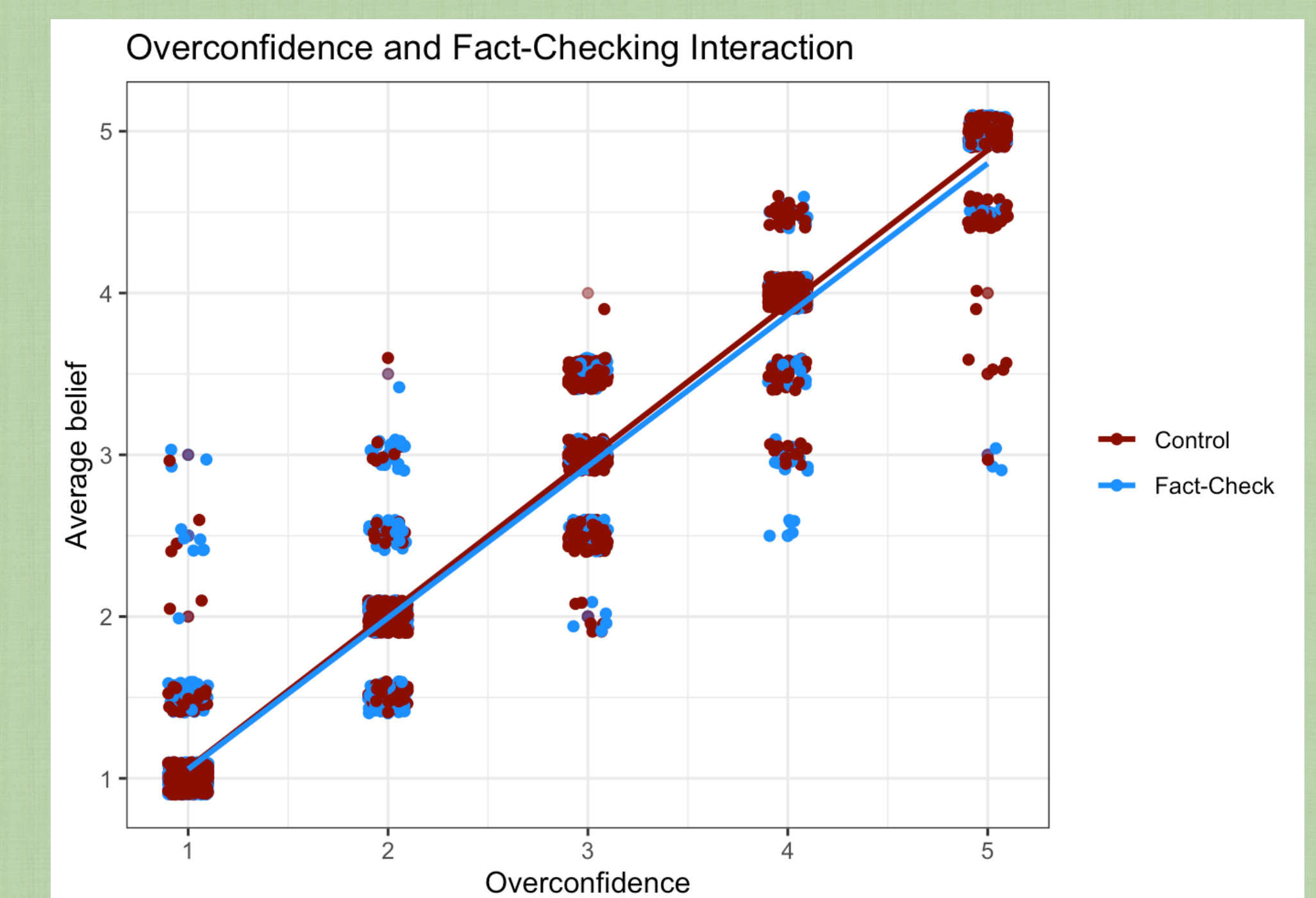
IV = GOT

Belief ~ GOT * Fact-check + (1 | item) + (1 | subj)

GOT [b = .04, p = .005]

Fact-Check [b = -1.01, p < .001]

GOT : Fact-Check [b = .06, p = .034]



TAKE HOME POINTS

- Overconfidence appears to moderate the efficacy of psychological inoculation and fact-checking (albeit modestly)
- Given overconfidence is associated with increased belief and sharing of misinformation¹, those more vulnerable may not receive the intended benefits

1. See Lyons et al., 2021; Pennycook et al., 2024 Vranic et al., 2021

Future work could examine other interventions commonly used to see whether this effect persists. Additionally, considerations for other cognitive biases would be beneficial to ensure the intended effects. **Limitations** include convenience sampling – representativeness and selection biases – and temporal effects.